

| Equation of Motion Time Discretization

| Newton's 2nd Law as a first-order system

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{v} \\ \mathbf{M}\dot{\mathbf{v}} = \mathbf{f}(\mathbf{x}, \mathbf{v}) \end{cases}$$

BDF1

$$\dot{\mathbf{y}}_t \approx \frac{\mathbf{y}_{t+1} - \mathbf{y}_t}{h}$$

| Explicit Euler

$$\begin{cases} \mathbf{x}^+ = \mathbf{x}^- + h \mathbf{v}^- \\ \mathbf{v}^+ = \mathbf{v}^- + h \mathbf{M}^{-1} \mathbf{f}(\mathbf{x}^-, \mathbf{v}^-) \end{cases}$$

| Implicit Euler

$$\begin{cases} \mathbf{x}^+ = \mathbf{x}^- + h \mathbf{v}^+ \\ \mathbf{M}(\mathbf{v}^+ - \mathbf{v}^-) = h \mathbf{f}(\mathbf{x}^+, \mathbf{v}^+) \end{cases}$$

| Implicit Euler

$$\begin{cases} \mathbf{x}^+ = \mathbf{x}^- + h \mathbf{v}^+ \\ \mathbf{M}(\mathbf{v}^+ - \mathbf{v}^-) = h \mathbf{f}(\mathbf{x}^+, \mathbf{v}^+) \end{cases}$$



$$\mathbf{r}(\mathbf{x}^+) = \frac{1}{h^2} \mathbf{M}(\mathbf{x}^+ - \tilde{\mathbf{x}}) - \mathbf{f}(\mathbf{x}^+) = \mathbf{0}$$



**Non-Linear
System of
Equations**

$$\frac{1}{h^2} \begin{bmatrix} m_1 & & & \\ & m_2 & & \\ & & \ddots & \\ & & & m_n \end{bmatrix} \begin{bmatrix} \mathbf{x}_1^+ - \tilde{\mathbf{x}}_1 \\ \mathbf{x}_2^+ - \tilde{\mathbf{x}}_2 \\ \vdots \\ \mathbf{x}_n^+ - \tilde{\mathbf{x}}_n \end{bmatrix} - \begin{bmatrix} \mathbf{f}_1(\mathbf{x}^+) \\ \mathbf{f}_2(\mathbf{x}^+) \\ \vdots \\ \mathbf{f}_n(\mathbf{x}^+) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

| Implicit Euler

$$\mathbf{r}(\mathbf{x}^+) = \frac{1}{h^2} \mathbf{M} (\mathbf{x}^+ - \tilde{\mathbf{x}}) - \mathbf{f}(\mathbf{x}^+) = \mathbf{0}$$

Unknown

Timestep Length

$\mathbf{x}^- + h \mathbf{v}^-$

Force: Non-Linear Function

Diagonal Mass Matrix: $\begin{bmatrix} m_1 & & \\ & \ddots & \\ & & m_n \end{bmatrix}$

The diagram illustrates the Implicit Euler method equation: $\mathbf{r}(\mathbf{x}^+) = \frac{1}{h^2} \mathbf{M} (\mathbf{x}^+ - \tilde{\mathbf{x}}) - \mathbf{f}(\mathbf{x}^+) = \mathbf{0}$. Annotations include: 'Unknown' pointing to $\mathbf{x}^+$; 'Timestep Length' pointing to h ; $\mathbf{x}^- + h \mathbf{v}^-$ pointing to $\tilde{\mathbf{x}}$; 'Force: Non-Linear Function' pointing to $\mathbf{f}(\mathbf{x}^+)$; and 'Diagonal Mass Matrix' pointing to $\mathbf{M}$, which is represented as a diagonal matrix with elements $m_1, \dots, m_n$.

| Newton-Raphson Method

We have:

$$\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

We want:

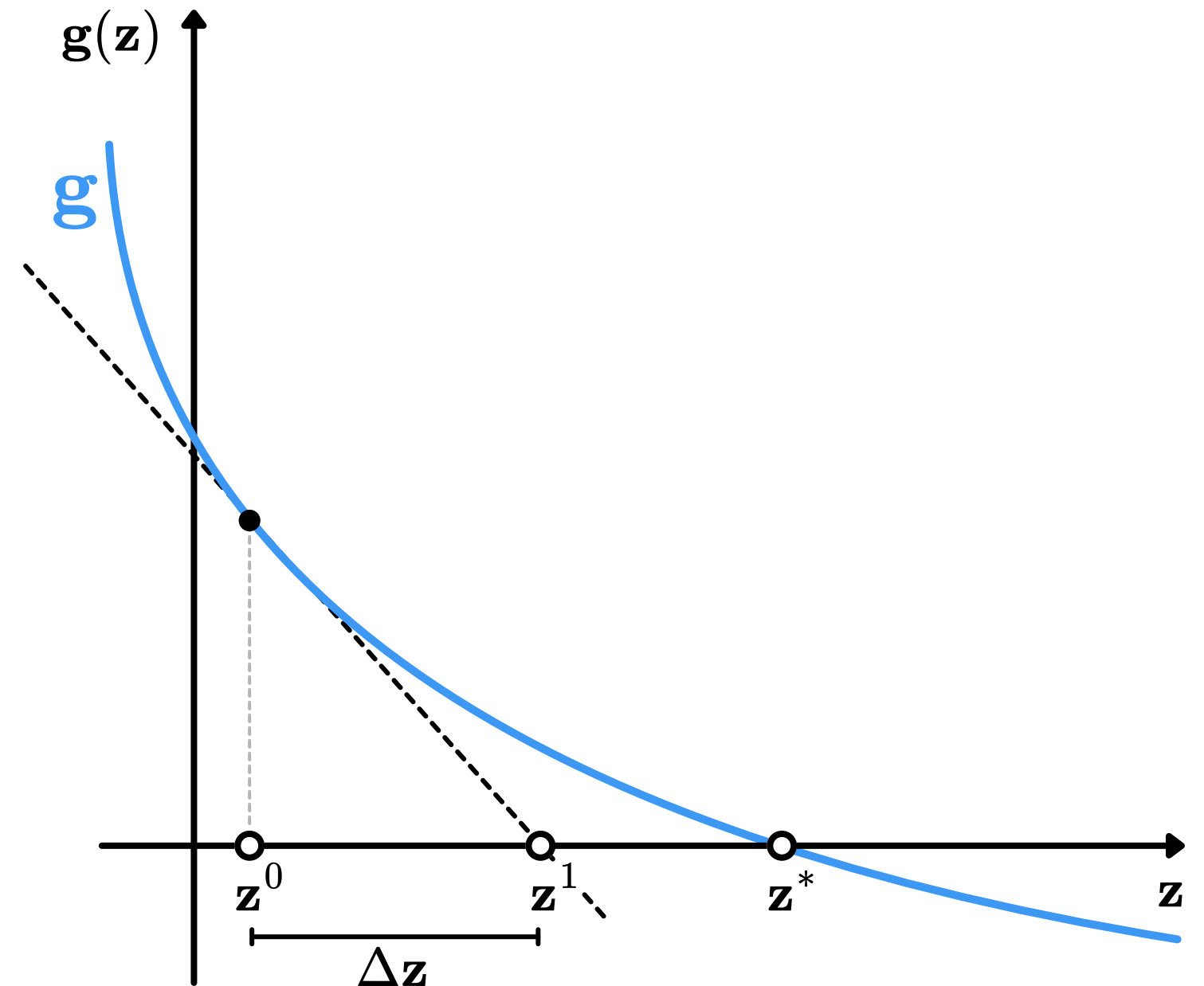
$$\mathbf{g}(\mathbf{z}^*) = \mathbf{0}$$

Linearize around a guess:

$$\mathbf{g}(\mathbf{z}^{(k)} + \Delta \mathbf{z}) \approx \mathbf{g}(\mathbf{z}^{(k)}) + \frac{\partial \mathbf{g}}{\partial \mathbf{z}}(\mathbf{z}^{(k)}) \Delta \mathbf{z} = \mathbf{0}$$

Linear System of Equations

$$\frac{\partial \mathbf{g}}{\partial \mathbf{z}} \Delta \mathbf{z} = -\mathbf{g}(\mathbf{z}^{(k)})$$
$$\mathbf{z}^{(k+1)} = \mathbf{z}^{(k)} + \Delta \mathbf{z}$$



| Newton-Raphson Method for Implicit Dynamic System

We have:

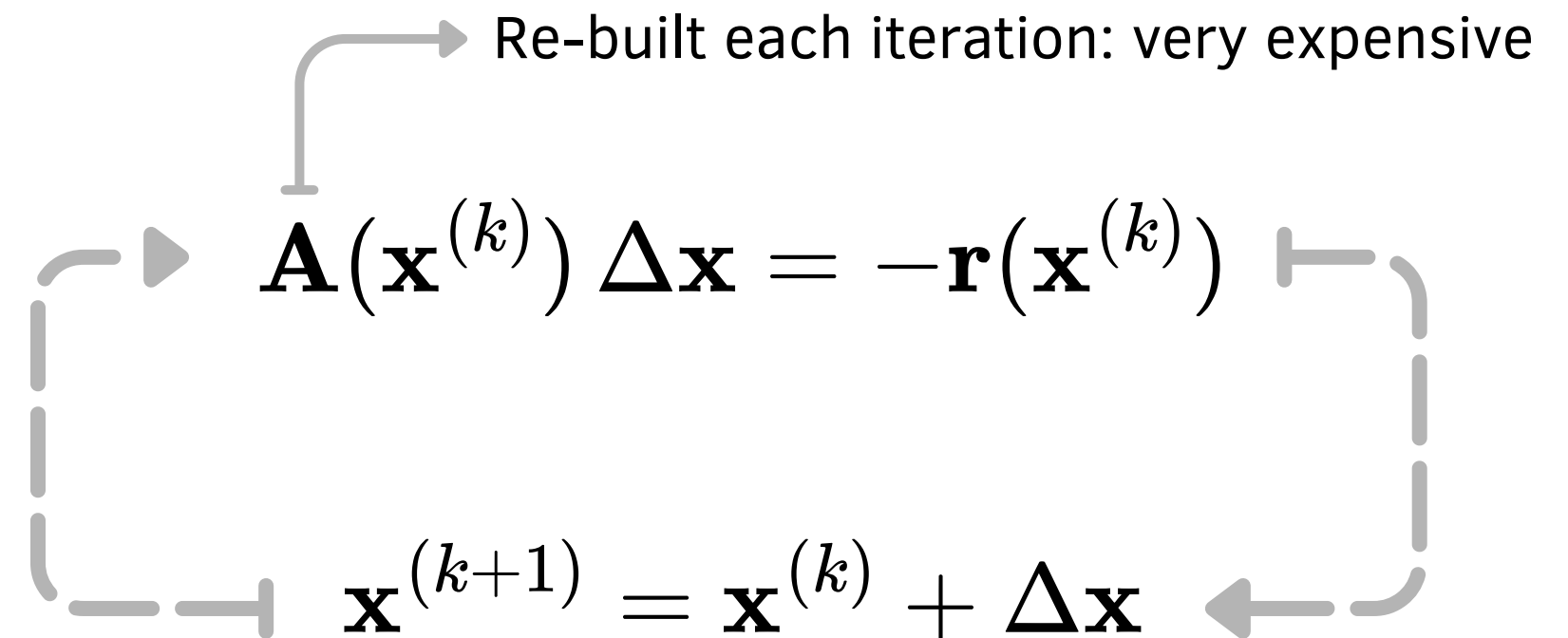
$$\mathbf{r} : \mathbb{R}^{3n} \rightarrow \mathbb{R}^{3n} \quad \text{such that:} \quad \mathbf{r}(\mathbf{x}) = \frac{1}{h^2} \mathbf{M}(\mathbf{x} - \tilde{\mathbf{x}}) - \mathbf{f}(\mathbf{x})$$

We want:

$$\mathbf{r}(\mathbf{x}^+) = \mathbf{0}$$

Linearize around a guess:

$$\frac{\partial \mathbf{r}}{\partial \mathbf{x}} = \underbrace{\frac{1}{h^2} \mathbf{M} - \frac{\partial \mathbf{f}}{\partial \mathbf{x}}}_{\mathbf{A} \in \mathbb{R}^{3n \times 3n}}$$



| Newton-Raphson Method Single Iteration

Expanded:

$$\underbrace{\left[\mathbf{M} - h^2 \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right]}_{\mathbf{A}(\mathbf{x}^-)} \Delta \mathbf{x} = \underbrace{h^2 \mathbf{f}(\mathbf{x}^-) + h \mathbf{M} \mathbf{v}^-}_{\mathbf{b}(\mathbf{x}^-)}$$

Algorithm:

- 1 **build:** $\mathbf{A}, \mathbf{b}$
- 2 **solve:** $\mathbf{A} \Delta \mathbf{x} = \mathbf{b}$
- 3 $\mathbf{x}^+ = \mathbf{x}^- + \Delta \mathbf{x}$
- 4 $\mathbf{v}^+ = \frac{1}{h} (\mathbf{x}^+ - \mathbf{x}^-)$

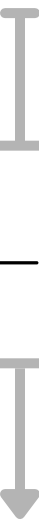
| Newton-Raphson Method Single Iteration

$$\left[\mathbf{M} - h \frac{\partial \mathbf{f}}{\partial \mathbf{v}} - h^2 \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right] \Delta \mathbf{x} = h \left[h \mathbf{f}(\mathbf{x}^-, \mathbf{v}^-) + \left(\mathbf{M} - h \frac{\partial \mathbf{f}}{\partial \mathbf{v}} \right) \mathbf{v}^- \right]$$

| Recasting Root-Finding as Minimization

Does exist a: $g(\mathbf{x})$ such that: $\nabla g(\mathbf{x}) = \mathbf{r}(\mathbf{x})$

$$g(\mathbf{x}) = \frac{1}{2h^2} (\mathbf{x} - \tilde{\mathbf{x}})^\top \mathbf{M} (\mathbf{x} - \tilde{\mathbf{x}}) + U(\mathbf{x})$$

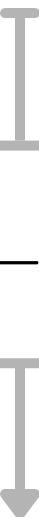

$$\mathbf{f}(\mathbf{x}) = -\frac{\partial U}{\partial \mathbf{x}}(\mathbf{x}) \quad \text{Conservative Forces}$$

$$\nabla g(\mathbf{x}) = \frac{1}{h^2} \mathbf{M} (\mathbf{x} - \tilde{\mathbf{x}}) - \mathbf{f}(\mathbf{x}) = \mathbf{r}(\mathbf{x})$$

| Recasting Root-Finding as Minimization

Does exists a: $g(\mathbf{x})$ such that: $\nabla g(\mathbf{x}) = \mathbf{r}(\mathbf{x})$

$$g(\mathbf{x}) = \frac{1}{2h^2} (\mathbf{x} - \tilde{\mathbf{x}})^\top \mathbf{M} (\mathbf{x} - \tilde{\mathbf{x}}) + U(\mathbf{x})$$



$\mathbf{f}(\mathbf{x}) = -\frac{\partial U}{\partial \mathbf{x}}(\mathbf{x})$

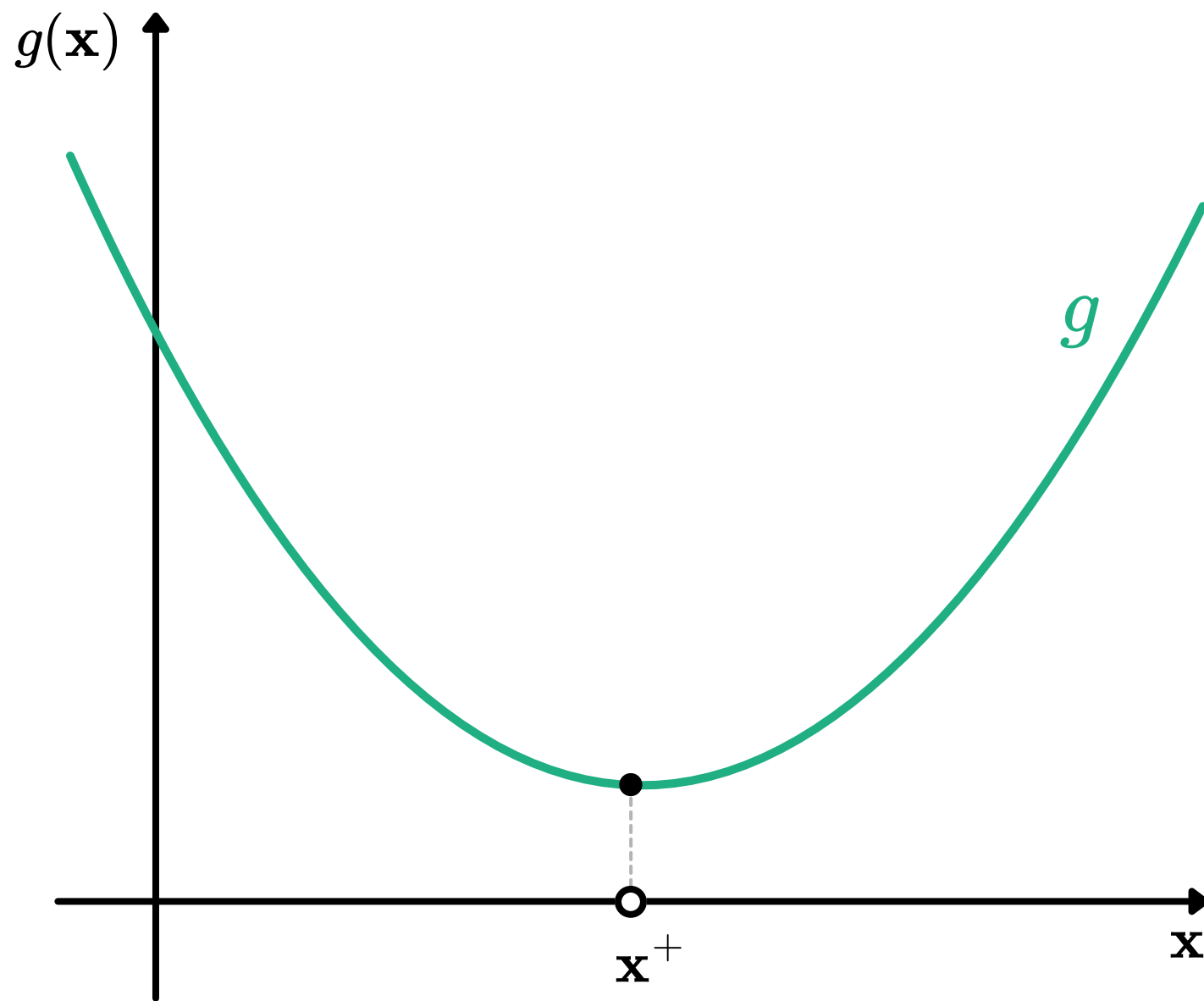
Conservative Forces

$$\mathbf{r}(\mathbf{x}) = \mathbf{0} \quad \Longleftrightarrow \quad \nabla g(\mathbf{x}) = \mathbf{0} \quad \begin{array}{c} \text{if Convex} \\ \Longleftrightarrow \end{array} \quad \min_{\mathbf{x}} g(\mathbf{x})$$

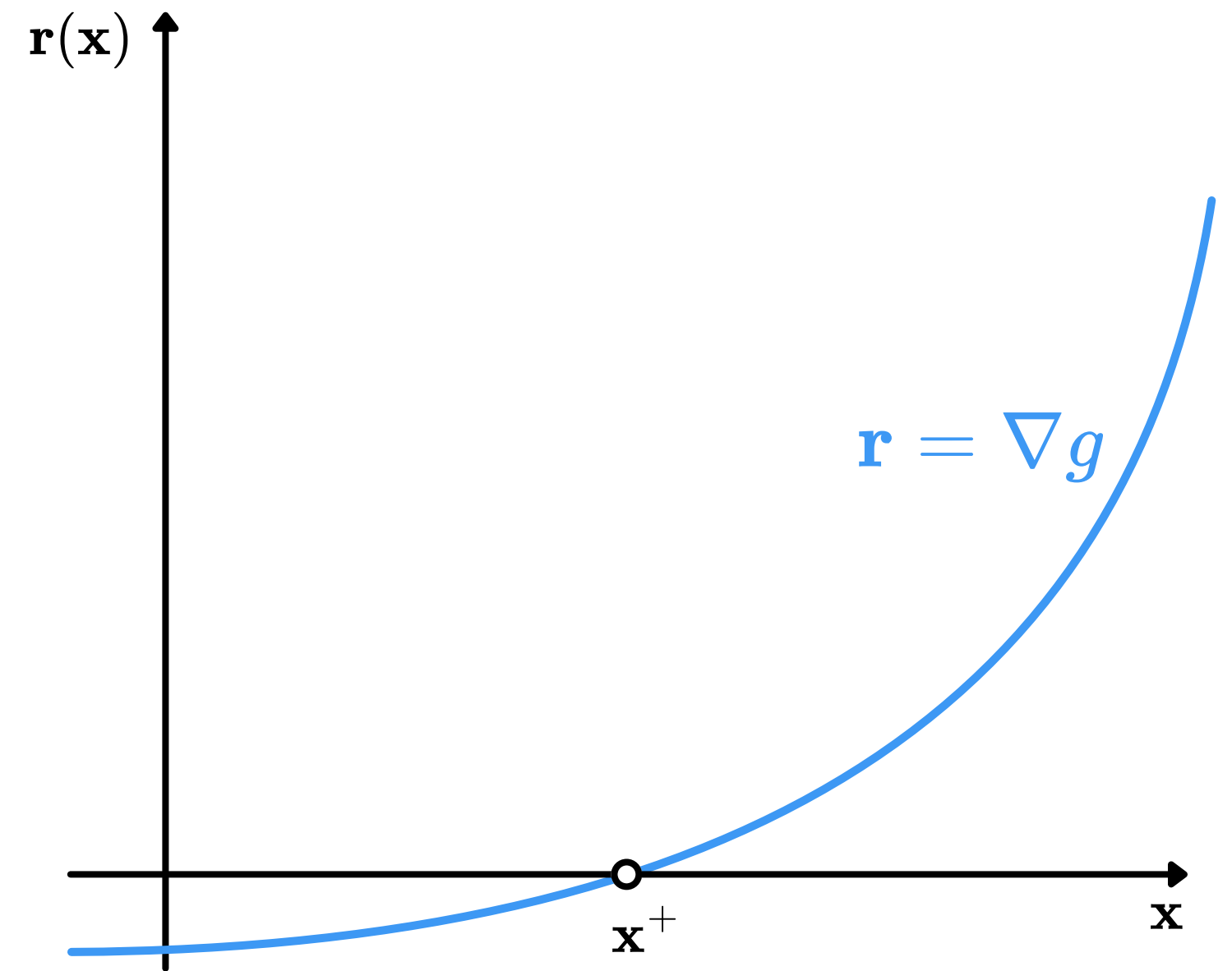
| Root-Finding and Minimization Equivalence

Equivalent Problems:

$$\mathbf{x}^+ = \arg \min_{\mathbf{x}} g(\mathbf{x})$$



$$\mathbf{r}(\mathbf{x}^+) = 0$$



| Descent Methods How to Solve the Minimization Problem

```
1 Initialize  $\mathbf{x}^{(0)}$ 
2 for  $k = 0, 1, \dots, N - 1$  do
3      $\Delta \mathbf{x} = \text{DescentDirection}()$ 
4      $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \cdot \Delta \mathbf{x}$ 
5  $\mathbf{x}^+ = \mathbf{x}^{(N)}$ 
6  $\mathbf{v}^+ = \frac{1}{h}(\mathbf{x}^+ - \mathbf{x}^-)$ 
```

| Jacobian:

$$\mathbf{J} = \nabla g(\mathbf{x}) = \mathbf{r}(\mathbf{x}) \in \mathbb{R}^{3n}$$

| Hessian:

$$\mathbf{H} = \nabla^2 g(\mathbf{x}) = \mathbf{A}(\mathbf{x}) \in \mathbb{R}^{3n \times 3n}$$

| Gradient Descent:

$$\Delta \mathbf{x}^{(k)} = -\mathbf{J}^{(k)}$$

| Newton:

$$\Delta \mathbf{x}^{(k)} = -[\mathbf{H}^{(k)}]^{-1} \mathbf{J}^{(k)}$$

| Quasi-Newton:

$$\Delta \mathbf{x}^{(k)} = -\mathbf{P}^{(k)} \mathbf{J}^{(k)}, \quad \mathbf{P}^{(k)} \approx [\mathbf{H}^{(k)}]^{-1}$$

| Projective Dynamics:

$$\Delta \mathbf{x}^{(k)} = -\mathbf{L}_{\text{PD}}^{-1} \mathbf{J}_{\text{PD}}^{(k)}$$

$$\mathbf{J}_{\text{PD}} = \nabla g_{\text{PD}}(\mathbf{x}), \quad \mathbf{L}_{\text{PD}} \approx \mathbf{H} \text{ (fixed)}$$

| Baraff-Witkin:

$$\Delta \mathbf{x} = -\mathbf{H}^{-1} \mathbf{J}, \quad k = 1$$

| Newton's Minimization Method = Newton-Raphson Root Finding's Method

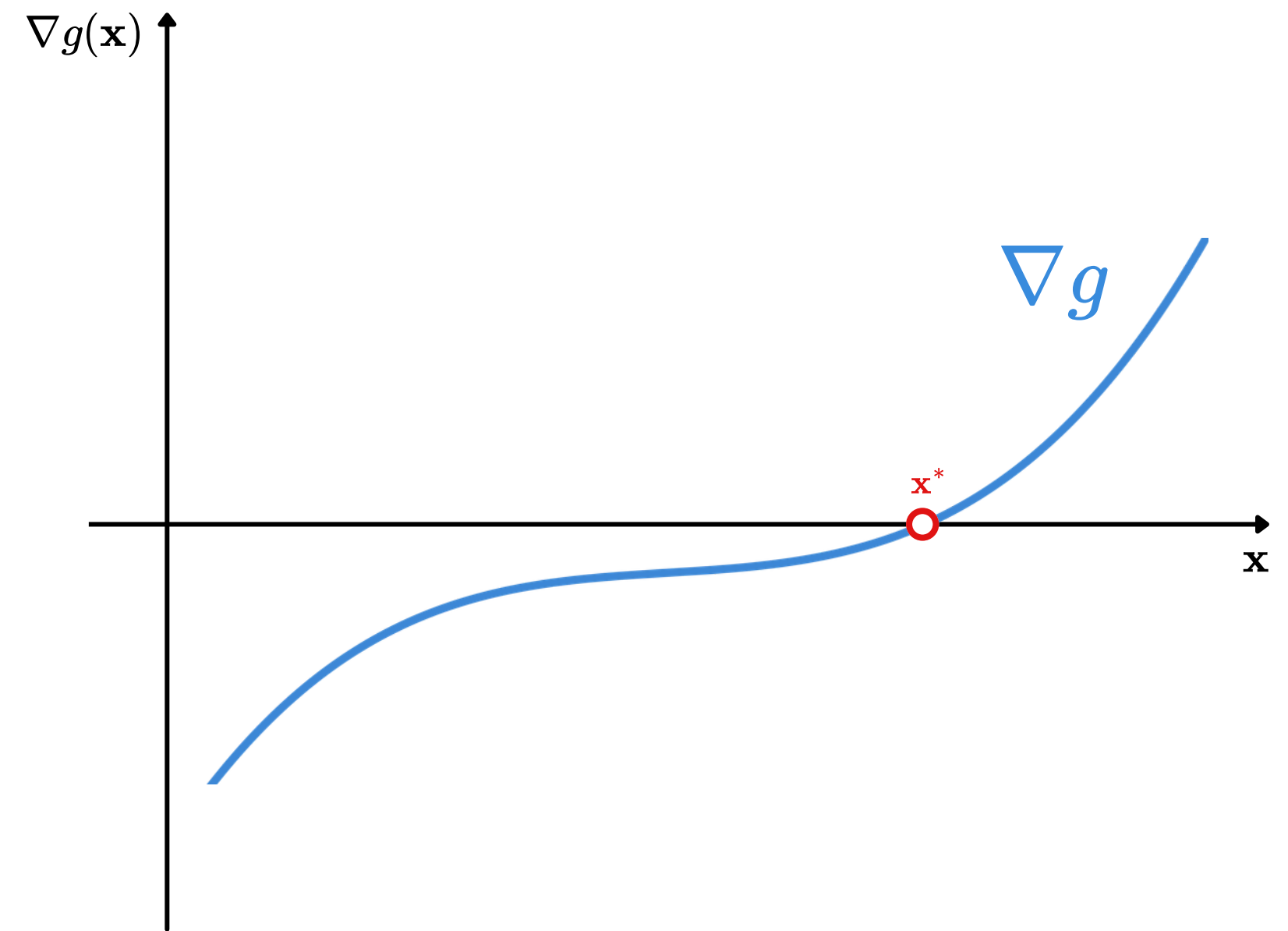
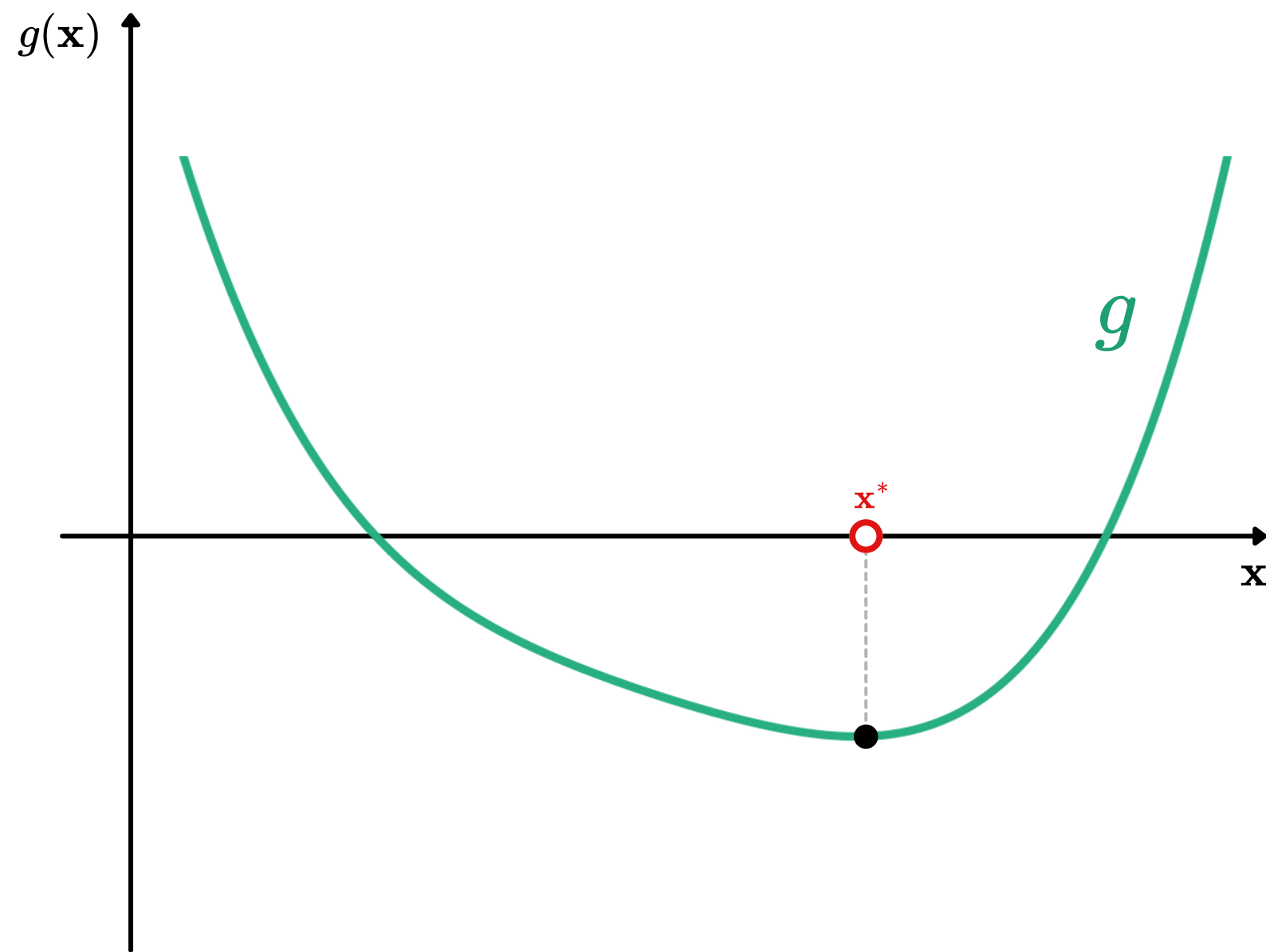
function NEWTONMETHOD($\mathbf{x}^-$, $\mathbf{v}^-$, N)

```
1 Initialize  $\mathbf{x}^{(0)}$ 
2 for  $k = 0, 1, \dots, N - 1$  do
3     BUILD  $\mathbf{J}^{(k)}$ ,  $\mathbf{H}^{(k)}$ 
4     SOLVE  $\mathbf{H}^{(k)} \Delta \mathbf{x} = -\mathbf{J}^{(k)}$ 
5      $\mathbf{x}^{(k+1)} \leftarrow \mathbf{x}^{(k)} + \Delta \mathbf{x}$ 
6  $\mathbf{x}^+ \leftarrow \mathbf{x}^{(N)}$ 
7  $\mathbf{v}^+ \leftarrow \frac{1}{h}(\mathbf{x}^+ - \mathbf{x}^-)$ 
```

function NEWTONRAPHSON($\mathbf{x}^-$, $\mathbf{v}^-$, N)

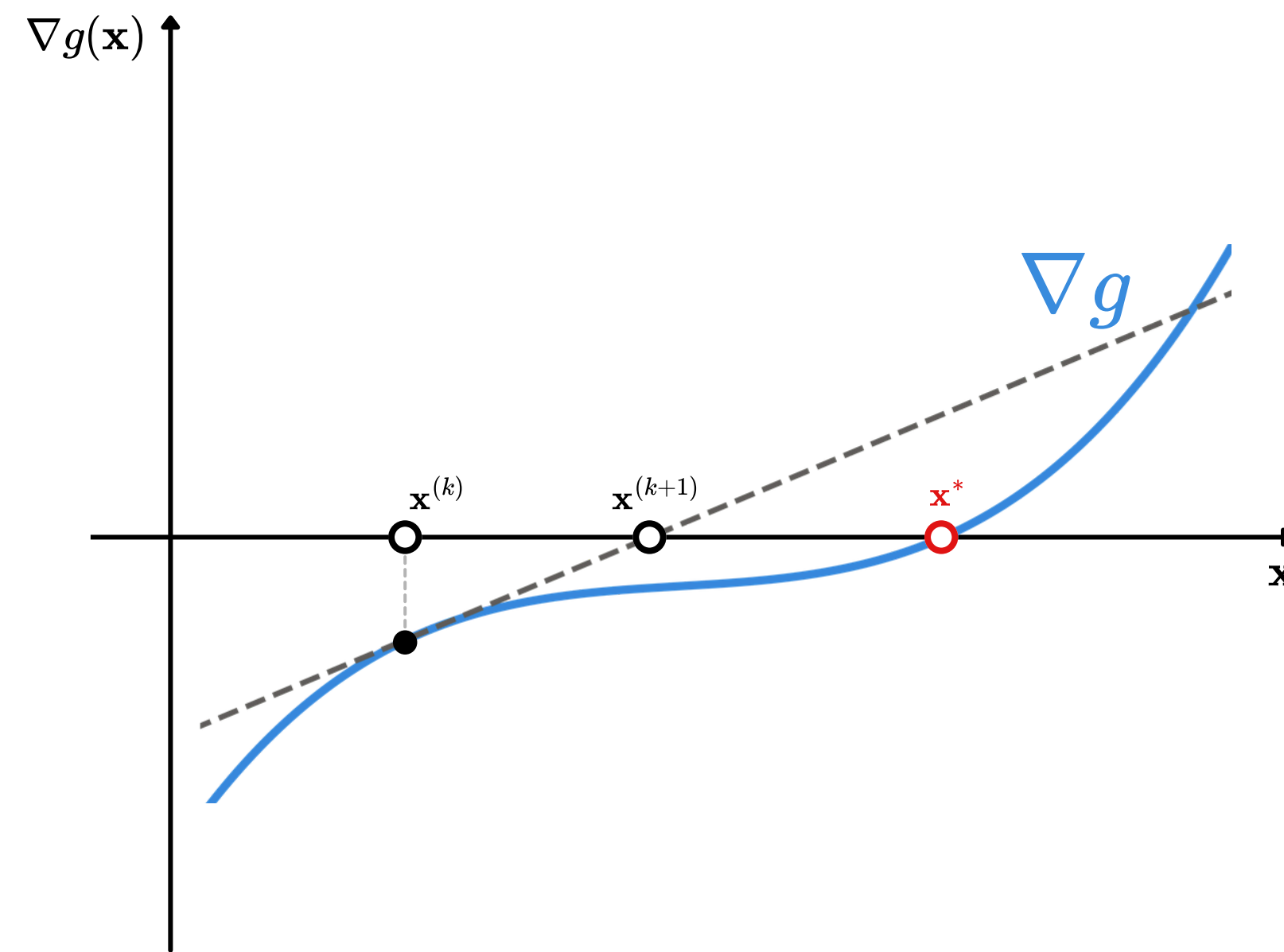
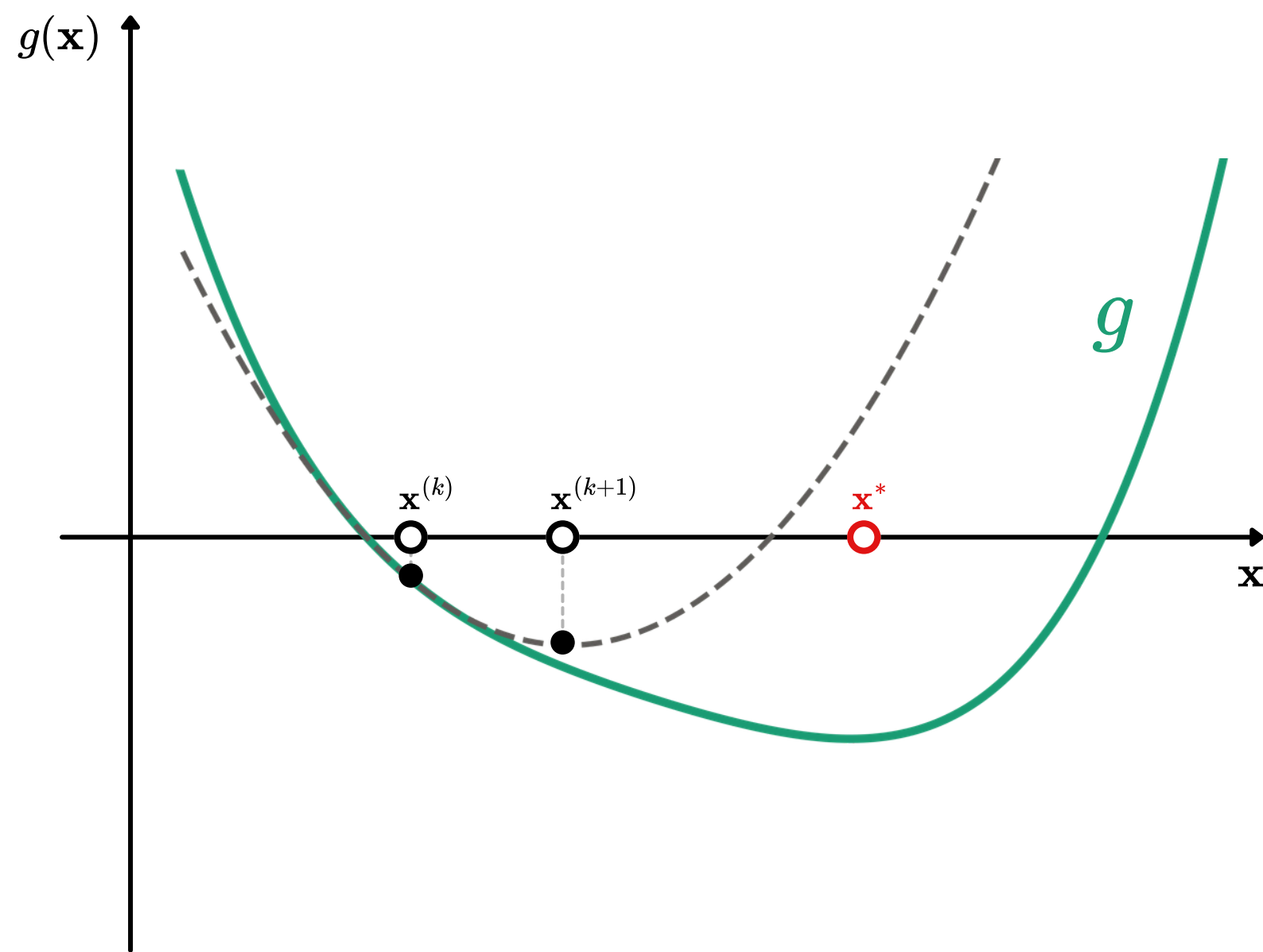
```
1 Initialize  $\mathbf{x}^{(0)}$ 
2 for  $k = 0, 1, \dots, N - 1$  do
3     BUILD  $\mathbf{A}^{(k)}$ ,  $\mathbf{b}^{(k)}$ 
4     SOLVE  $\mathbf{A}^{(k)} \Delta \mathbf{x} = \mathbf{b}^{(k)}$ 
5      $\mathbf{x}^{(k+1)} \leftarrow \mathbf{x}^{(k)} + \Delta \mathbf{x}$ 
6  $\mathbf{x}^+ \leftarrow \mathbf{x}^{(N)}$ 
7  $\mathbf{v}^+ \leftarrow \frac{1}{h}(\mathbf{x}^+ - \mathbf{x}^-)$ 
```

| Convexity and Newton's Method When the Energy **IS** Convex



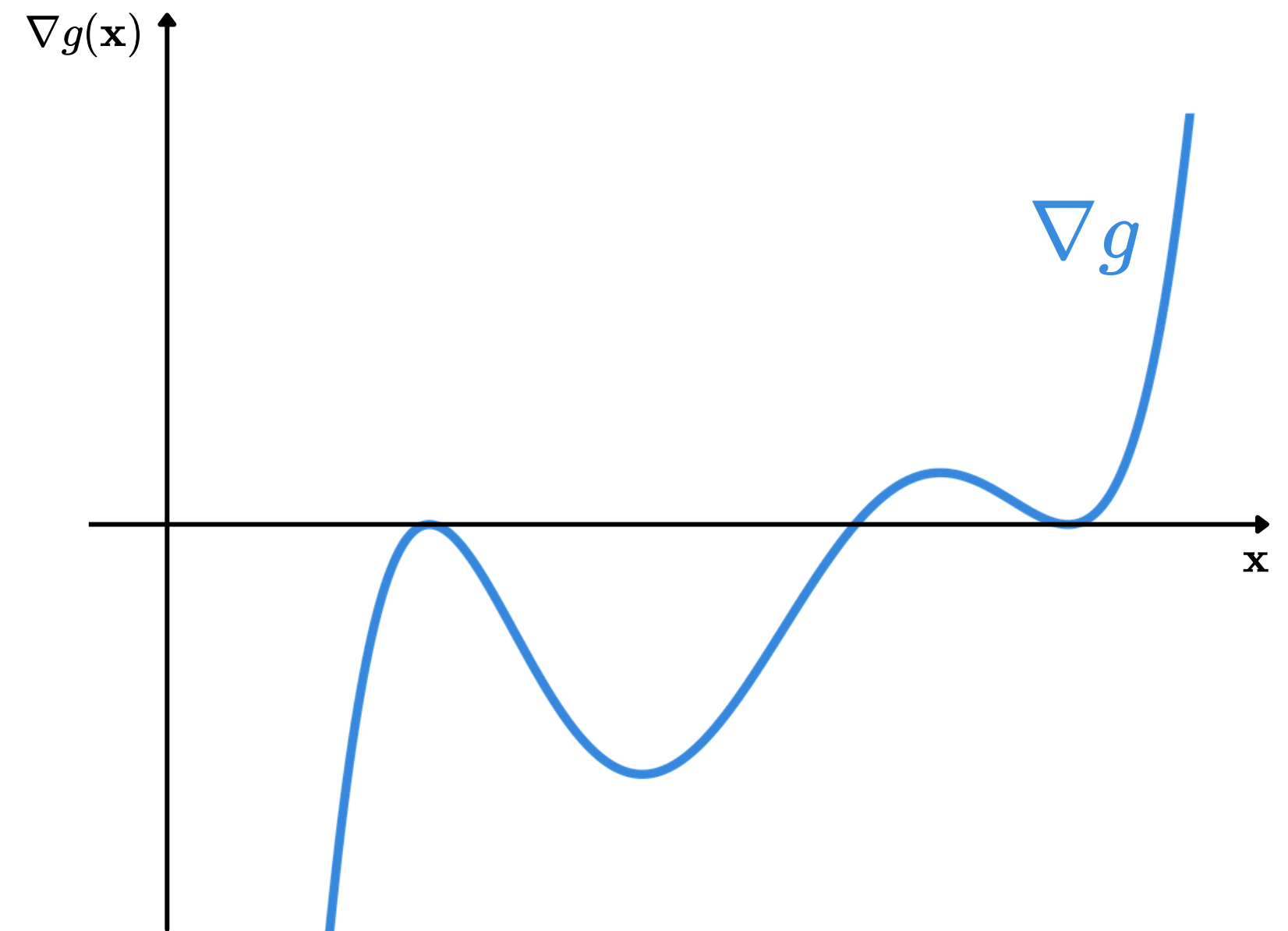
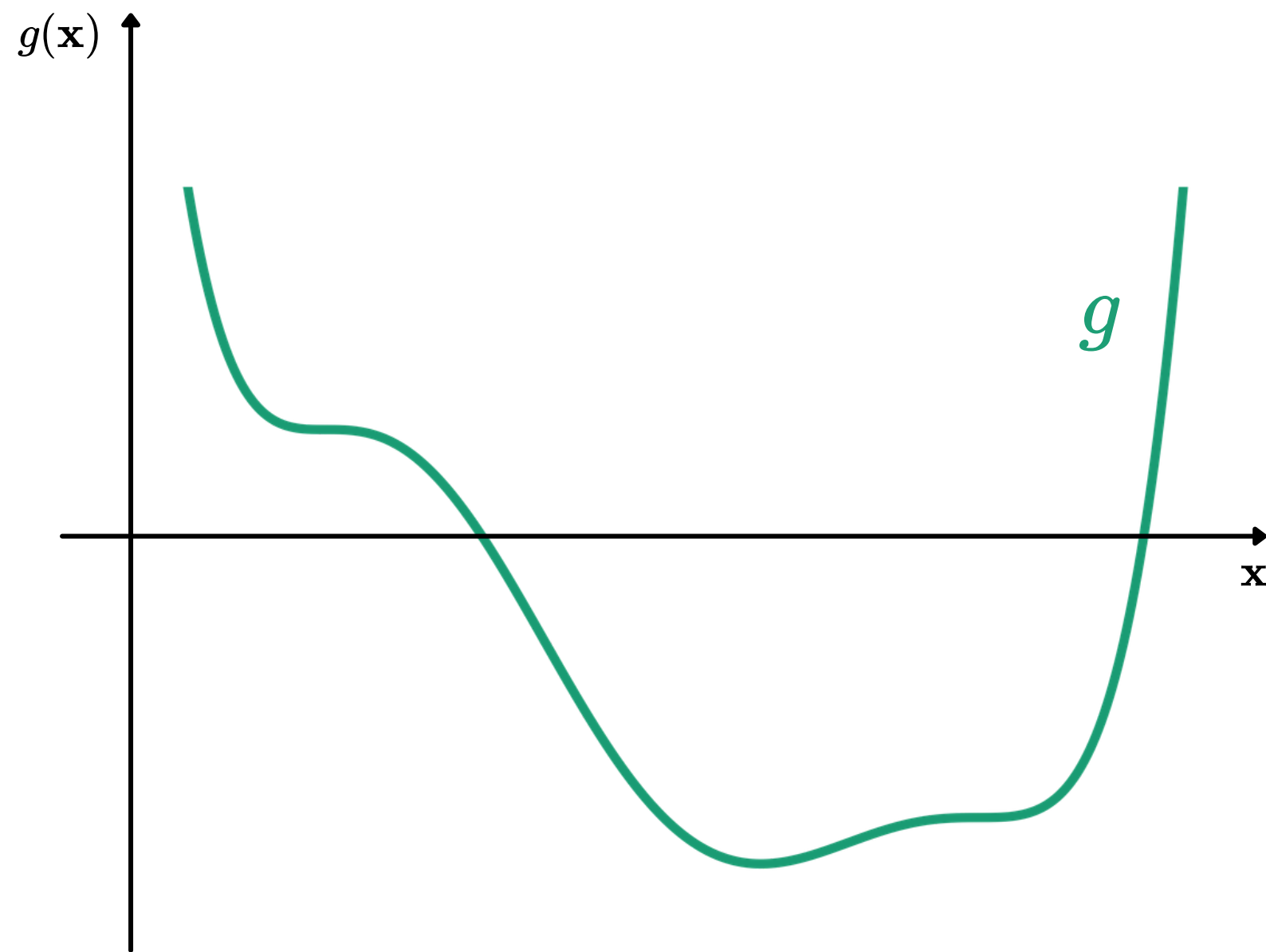
| Convexity and Newton's Method When the Energy **IS** Convex

$$g(\mathbf{x}^{(k+1)}) < g(\mathbf{x}^{(k)})$$

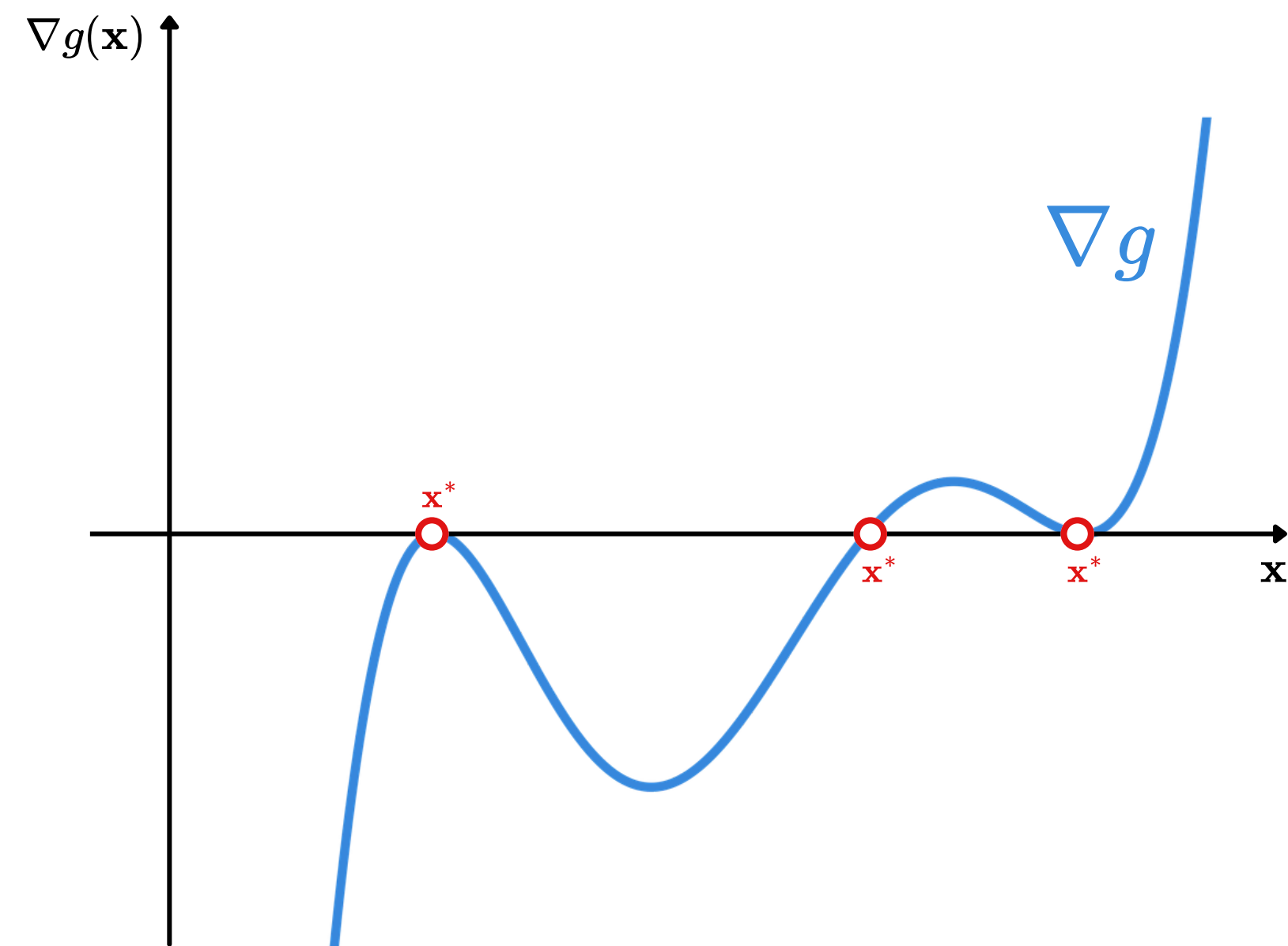
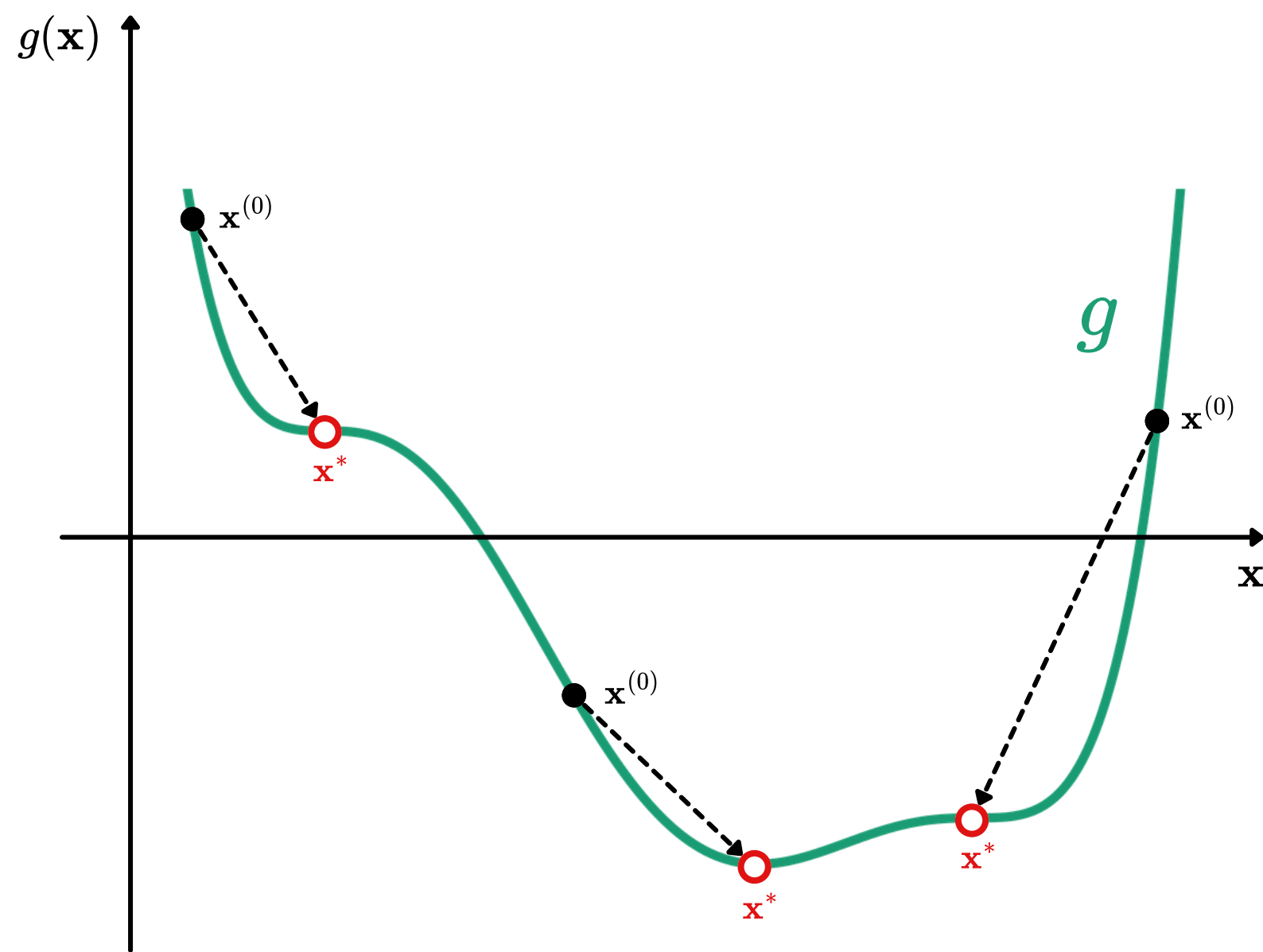


| Convexity and Newton's Method When the Energy **IS NOT** Convex

- There might be multiple stationary points
- There might be areas where the curvature is negative

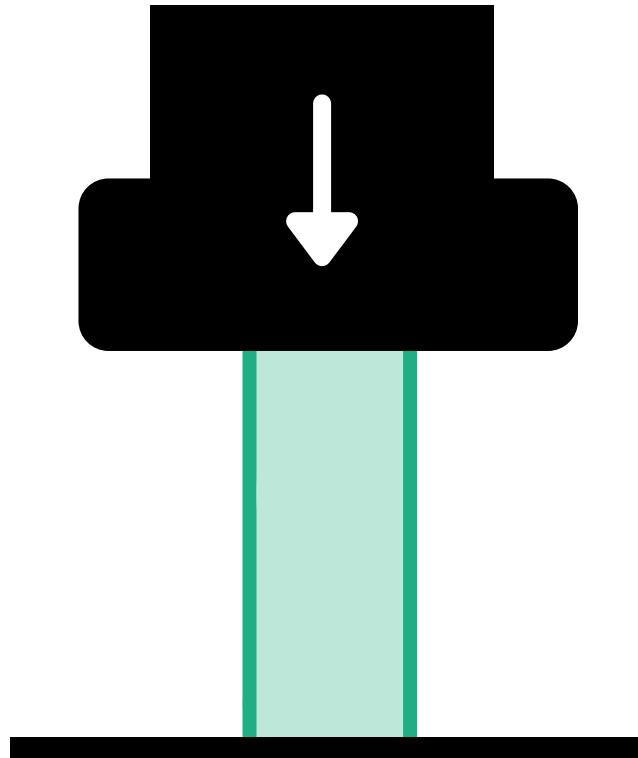


| Convexity and Newton's Method: Multiple Stationary Points

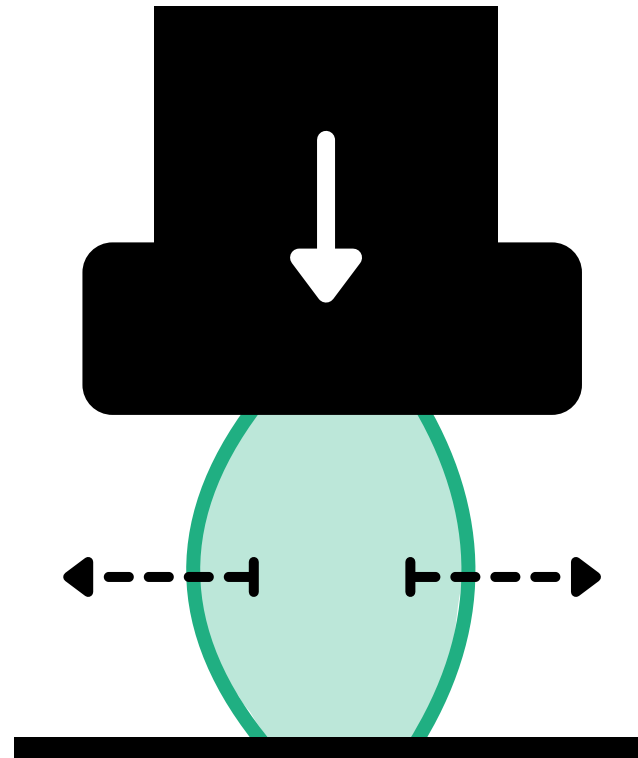


| Multiple Valid Energy States

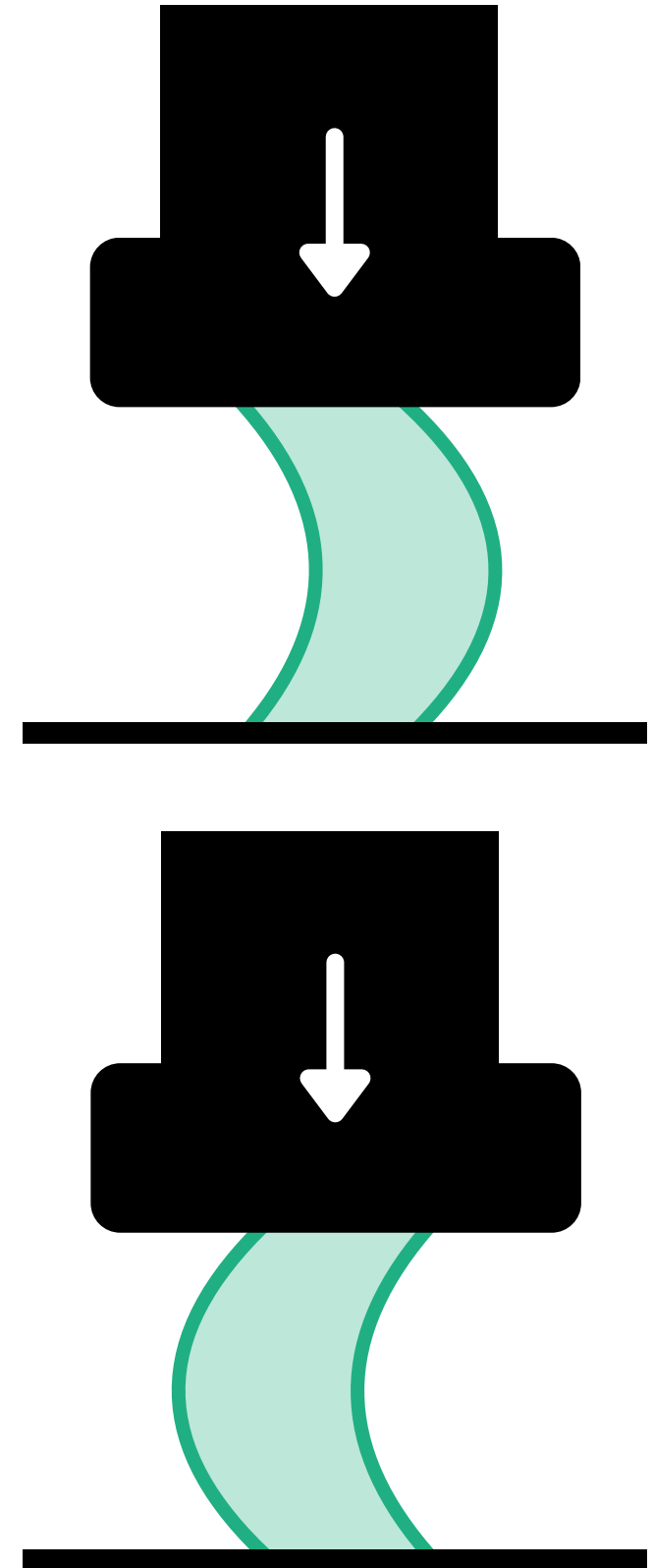
t_1



t_2



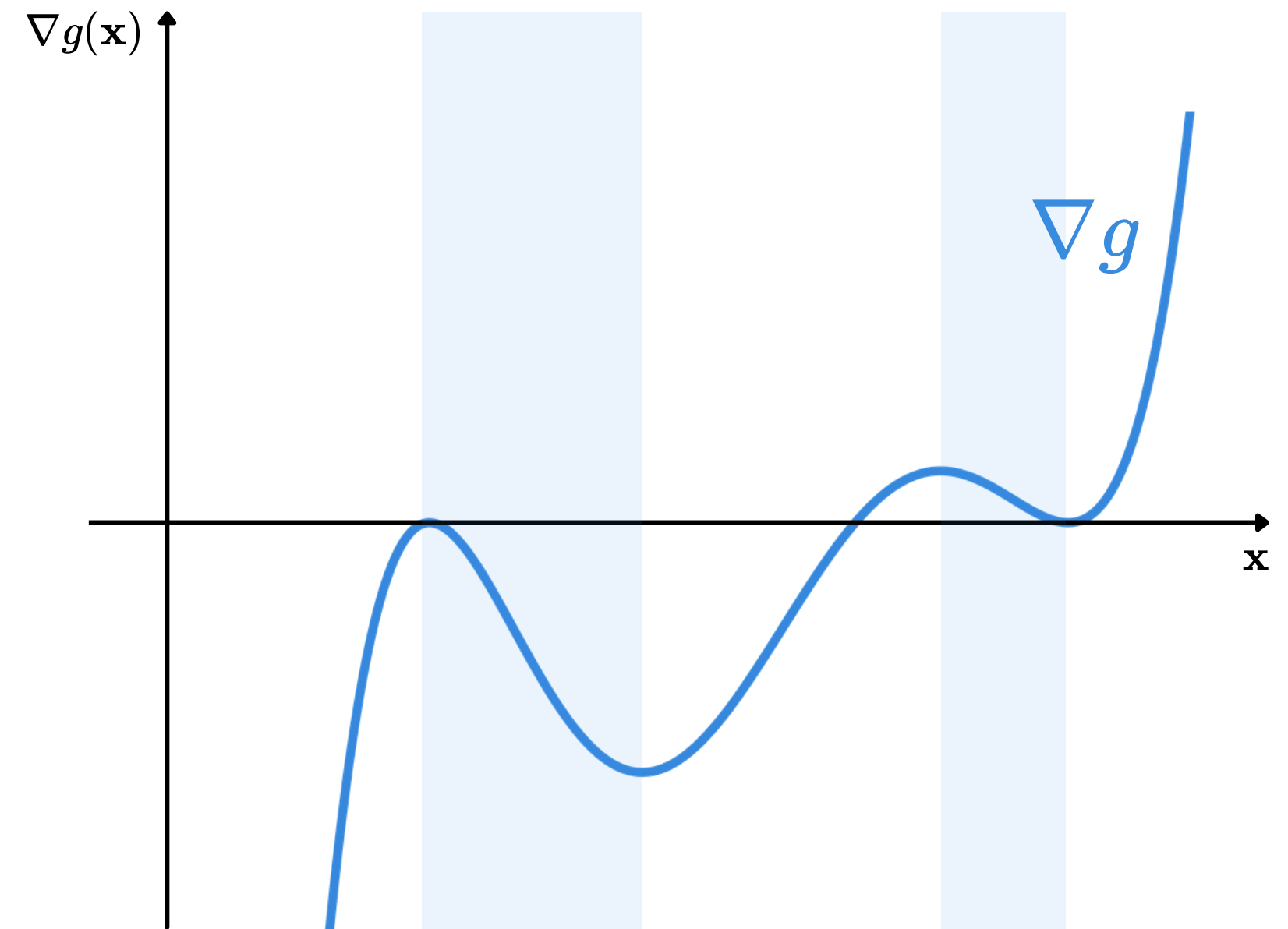
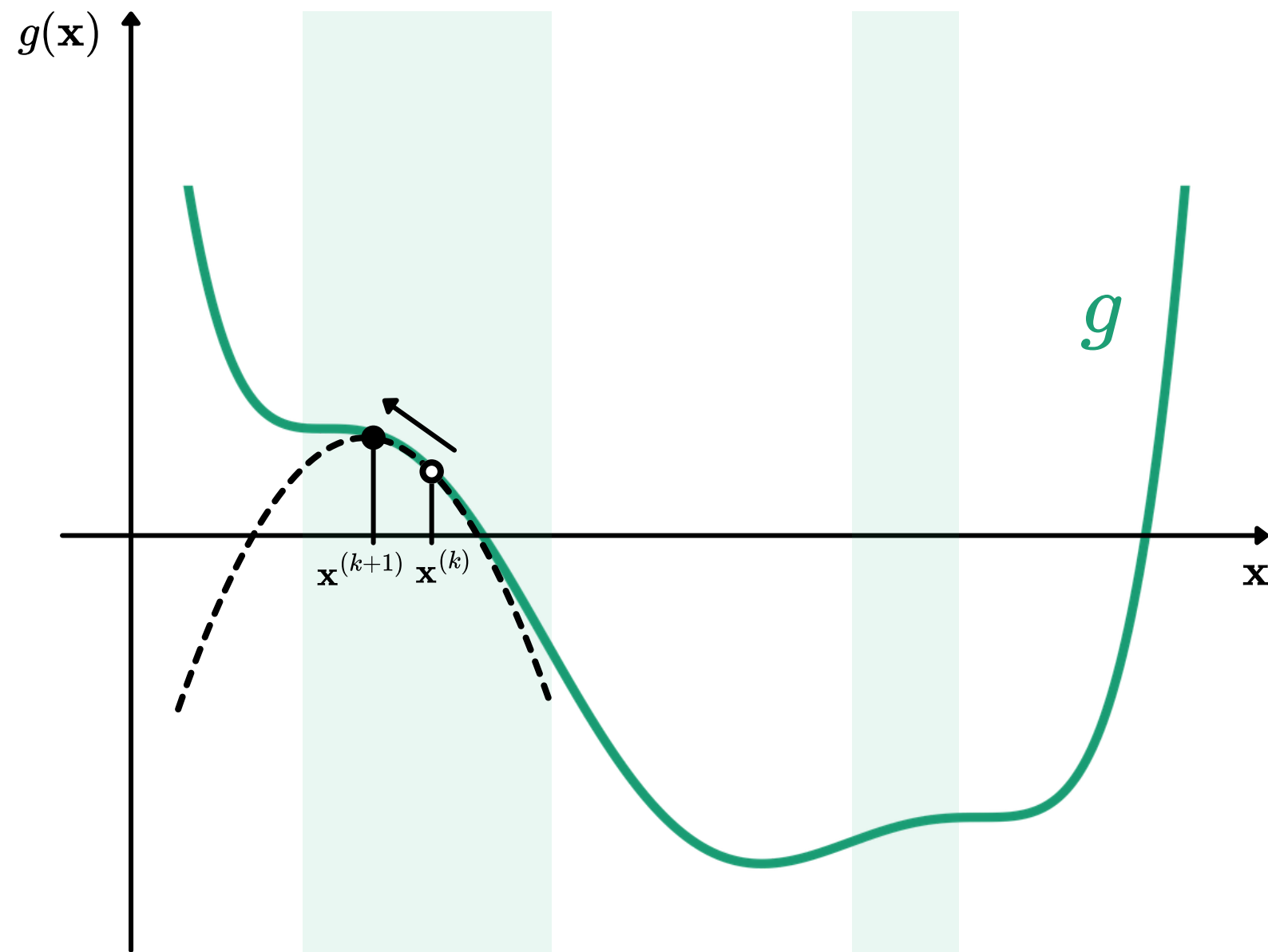
t_3



| Convexity and Newton's Method: Negative Curvature

If the curvature is negative, by taking the Newton's step, there might be an **increase** in Energy

$$g(\mathbf{x}^{(k+1)}) > g(\mathbf{x}^{(k)})$$



| Convexity and Newton's Method: Solve the Negative Curvature Problem

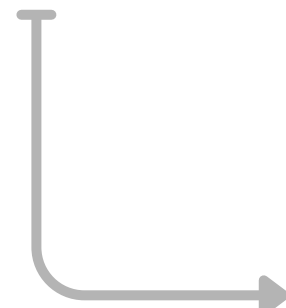
A matrix is SPD (**S**ymmetric **P**ositive **D**efinite) if it is symmetric and all its eigenvalues are positive:

$$\mathbf{A} = \mathbf{A}^\top \quad \text{and} \quad \mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top, \quad \lambda_i > 0 \quad \forall i.$$

and it is written: $\mathbf{A} \succ 0$

$\mathbf{H} \not\succ 0$: $\Delta \mathbf{x} = -\mathbf{H}^{-1} \mathbf{J}$ then $\Delta \mathbf{x}$ may not be a descent direction

$\bar{\mathbf{H}} \succ 0$: $\Delta \mathbf{x} = -\bar{\mathbf{H}}^{-1} \mathbf{J}$ then $\alpha \cdot \Delta \mathbf{x}$ is guaranteed to be a descent direction for a sufficiently small α

 is the SPD-Projection of $\mathbf{H}$

| Convexity and Newton's Method: Solve the Negative Curvature Problem

Always Convex

$$g(\mathbf{x}) = \frac{1}{2h^2} (\mathbf{x} - \tilde{\mathbf{x}})^\top \mathbf{M} (\mathbf{x} - \tilde{\mathbf{x}}) + U(\mathbf{x})$$

$\mathbf{H} =$

$$\frac{1}{h^2} \mathbf{M} + \nabla^2 U(\mathbf{x})$$

Always SPD Might not be SPD!

The diagram illustrates the components of the Hessian matrix $\mathbf{H}$ in Newton's method. The function $g(\mathbf{x})$ is composed of a quadratic term and a potential term $U(\mathbf{x})$. The quadratic term is always convex, and its Hessian $\frac{1}{h^2} \mathbf{M}$ is always symmetric positive definite (SPD). The potential term $U(\mathbf{x})$ may not be convex, and its Hessian $\nabla^2 U(\mathbf{x})$ might not be SPD, which is the negative curvature problem.